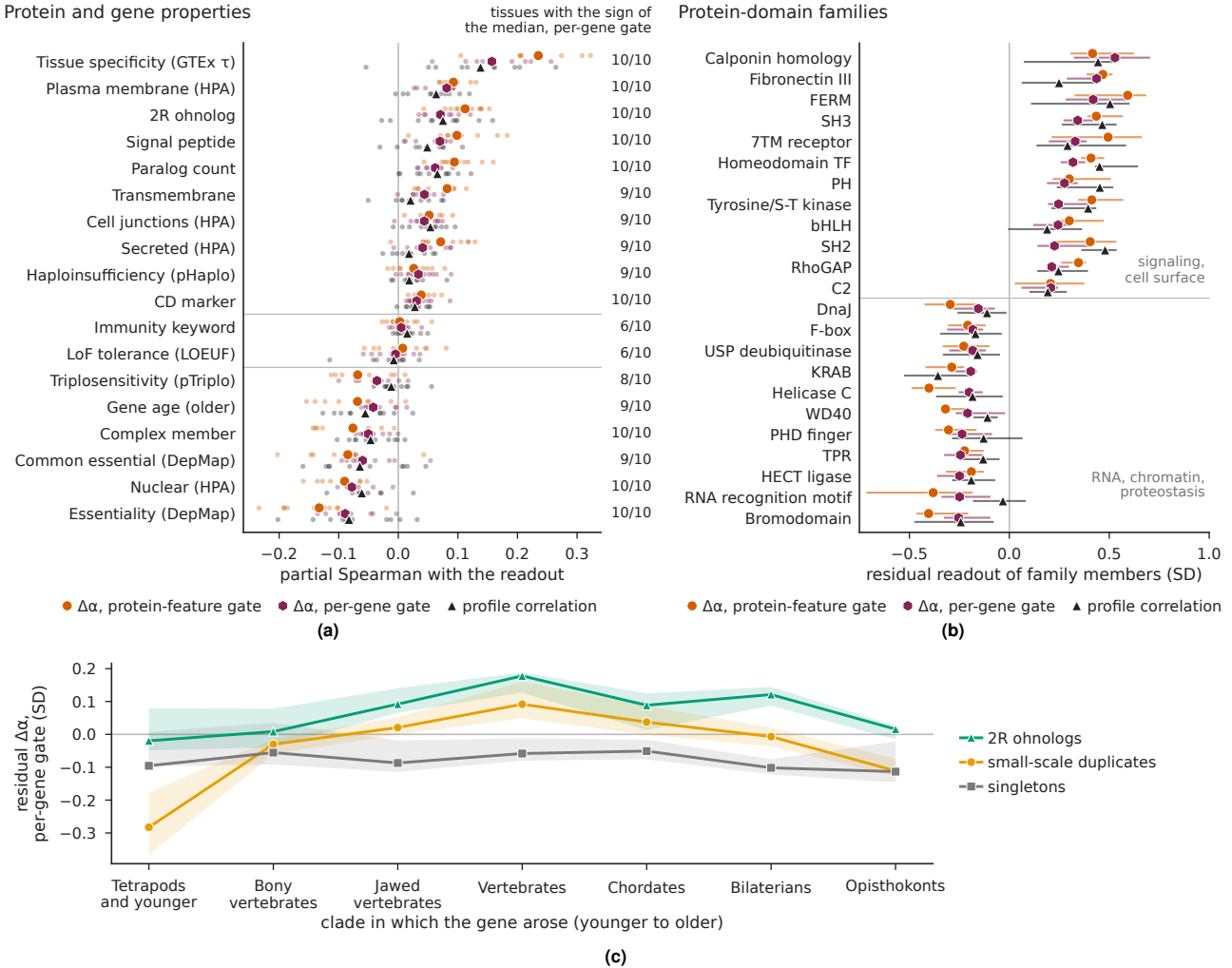

Protein and gene properties
tissues with the sign of the median, per-gene gate
Tissue specificity (GTEx τ)
Plasma membrane (HPA)
2R ohnolog
Signal peptide
Paralog count
Transmembrane
Cell junctions (HPA)
Secreted (HPA)
Haploinsufficiency (pHaplo)
CD marker
Immunity keyword
LoF tolerance (LOEUF)
Triplosensitivity (pTriplo)
Gene age (older)
Complex member
Common essential (DepMap)
Nuclear (HPA)
Essentiality (DepMap)
10/10
10/10
10/10
10/10
10/10
9/10
9/10
9/10
9/10
10/10
6/10
6/10
8/10
9/10
10/10
9/10
10/10
10/10
−0.2
−0.1
0.0
0.1
0.2
0.3
partial Spearman with the readout
Δα, protein-feature gate
Δα, per-gene gate
profile correlation
(a)
Protein-domain families
Calponin homology
Fibronectin III
FERM
SH3
7TM receptor
Homeodomain TF
PH
Tyrosine/S-T kinase
bHLH
SH2
RhoGAP
C2
DnaJ
F-box
USP deubiquitinase
KRAB
Helicase C
WD40
PHD finger
TPR
HECT ligase
RNA recognition motif
Bromodomain
signaling, cell surface
RNA, chromatin, proteostasis
−0.5
0.0
0.5
1.0
residual readout of family members (SD)
Δα, protein-feature gate
Δα, per-gene gate
profile correlation
(b)
residual Δα, per-gene gate (SD)
0.2
0.1
0.0
−0.1
−0.2
−0.3
2R ohnologs
small-scale duplicates
singletons
Tetrapods and younger
Bony vertebrates
Jawed vertebrates
Vertebrates
Chordates
Bilaterians
Opisthokonts
clade in which the gene arose (younger to older)
(c)